

Project Code

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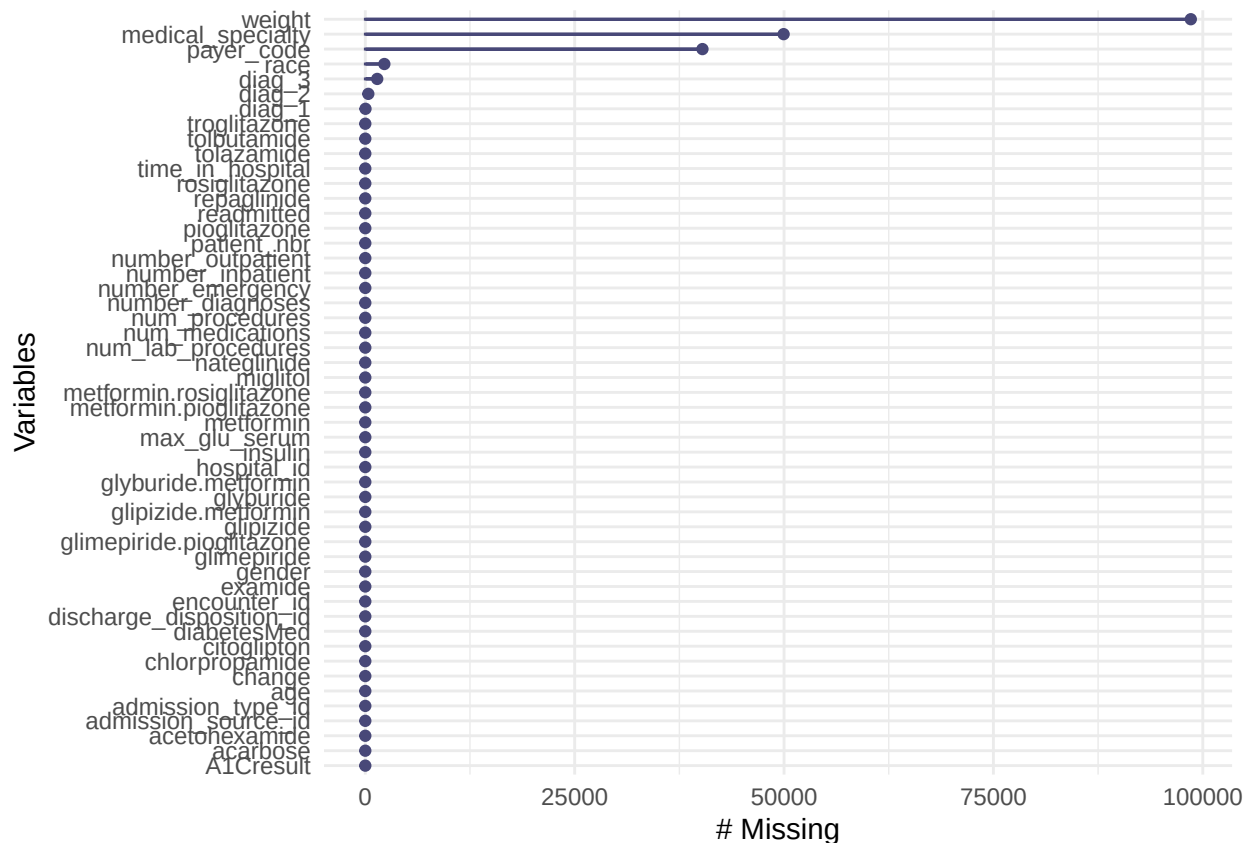
Research Question: “Which patient- or hospital-level factors are most predictive of readmission within 30 days among diabetic patients?”

Data Acquisition and Preprocessing

```
diabete <- read_csv("./diabetic_data_2.csv")  
#head(diabete, 3)
```

About **missingness** in data, three variables(payer_code, medical_specialty, weight) are dropped due to 10% or more missing values. Although they are direct or indirect related to the readmission outcome, too much missingness is very likely to make them ineffective predictor in the model. Now rows are dropped due to excessive missing values.

```
# Handle missingness  
diabete[diabete == "?"] <- NA  
gg_miss_var(diabete)
```



```
# Set missingness thresholds
col_na_thresh <- 0.1
row_na_thresh <- 0.1

# Filter out columns with >20% missing values
diabete <- diabete %>%
  dplyr::select(where(~ mean(is.na(.)) <= col_na_thresh))

# Filter out rows with >20% missing values
diabete <- diabete %>%
  filter(rowMeans(is.na(.)) <= row_na_thresh)

# Recode outcome variable
diabete <- diabete %>%
  mutate(readmission_binary = if_else(readmitted == "<30", 1, 0)) %>%
  mutate(factor(readmission_binary))
```

Exploratory Data Analysis

```
# Convert Age to ordered factor
age_levels <- c("[0-10)", "[10-20)", "[20-30)", "[30-40)", "[40-50)",
  "[50-60)", "[60-70)", "[70-80)", "[80-90)", "[90-100)")

diabete <- diabete |>
  mutate(age = factor(age, levels = age_levels, ordered = TRUE))
```

```
# Summarize key variables
```

```
key_vars <- diabetes %>%
  select(race, gender, age, admission_type_id, discharge_disposition_id,
         admission_source_id, A1Cresult, insulin,
         diabetesMed, readmission_binary, diag_1, time_in_hospital)
summary(key_vars)
```

```
##      race                gender          age      admission_type_id
## Length:101766      Length:101766      [70-80):26068      Min.       :1.000
## Class :character    Class :character    [60-70):22483      1st Qu.:1.000
## Mode  :character    Mode  :character    [50-60):17256      Median  :1.000
##                                     [80-90):17197      Mean    :2.024
##                                     [40-50): 9685      3rd Qu.:3.000
##                                     [30-40): 3775      Max.    :8.000
##                                     (Other): 5302
## discharge_disposition_id admission_source_id A1Cresult
## Min.       : 1.000           Min.       : 1.000      Length:101766
## 1st Qu.: 1.000           1st Qu.: 1.000      Class :character
## Median : 1.000           Median : 7.000      Mode  :character
## Mean    : 3.716           Mean    : 5.754
## 3rd Qu.: 4.000           3rd Qu.: 7.000
## Max.    :28.000           Max.    :25.000
##
##      insulin          diabetesMed      readmission_binary      diag_1
## Length:101766      Length:101766      Min.       :0.0000      Length:101766
## Class :character    Class :character    1st Qu.:0.0000      Class :character
## Mode  :character    Mode  :character    Median :0.0000      Mode  :character
##                                     Mean    :0.1116
##                                     3rd Qu.:0.0000
##                                     Max.    :1.0000
##
##      time_in_hospital
## Min.       : 1.000
## 1st Qu.: 2.000
## Median : 4.000
## Mean    : 4.396
## 3rd Qu.: 6.000
## Max.    :14.000
##
```

```
# Format reader freindly frequency table
```

```
key_vars_list <- c("race", "gender", "age", "admission_type_id", "discharge_disposition_id",
                  "admission_source_id", "A1Cresult",
                  "insulin", "diabetesMed", "readmission_binary", "time_in_hospital")
```

```
plots <- list()
```

```
for (var in key_vars_list) {
  cat("##", var)
  freq_table <- diabetes |>
    count(across(all_of(var)), name = "Count") |>
    mutate(Percent = round(Count / sum(Count) * 100, 1)) |>
    arrange(desc(Count)) |>
    rename(Level = 1)
```

```

print(kable(freq_table))
cat("\n\n")

# Bar plots
plot <- ggplot(freq_table, aes(x = fct_infreq(as.factor(Level)), y = Count)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(title = var, x = "Category", y = "Count") +
  theme_minimal(base_size = 8) +
  theme(axis.text.x = element_text(angle = 60, hjust = 1)) +
  theme(plot.title = element_text(hjust = 0.5))

plots[[var]] <- plot
}

```

```

## ## race
##
## |Level          | Count| Percent|
## |:-----:|-----:|-----:|
## |Caucasian      | 76099| 74.8|
## |AfricanAmerican| 19210| 18.9|
## |NA             | 2273 | 2.2|
## |Hispanic       | 2037 | 2.0|
## |Other          | 1506 | 1.5|
## |Asian          | 641  | 0.6|
##
##
## ## gender
##
## |Level          | Count| Percent|
## |:-----:|-----:|-----:|
## |Female         | 54708| 53.8|
## |Male           | 47055| 46.2|
## |Unknown/Invalid| 3    | 0.0|
##
##
## ## age
##
## |Level      | Count| Percent|
## |:-----:|-----:|-----:|
## |[70-80)    | 26068| 25.6|
## |[60-70)    | 22483| 22.1|
## |[50-60)    | 17256| 17.0|
## |[80-90)    | 17197| 16.9|
## |[40-50)    | 9685 | 9.5|
## |[30-40)    | 3775 | 3.7|
## |[90-100)   | 2793 | 2.7|
## |[20-30)    | 1657 | 1.6|
## |[10-20)    | 691  | 0.7|
## |[0-10)     | 161  | 0.2|
##
##
## ## admission_type_id
##

```

```

## | Level| Count| Percent|
## |-----:|-----:|-----:|
## |      1| 53990|    53.1|
## |      3| 18869|    18.5|
## |      2| 18480|    18.2|
## |      6|  5291|     5.2|
## |      5|  4785|     4.7|
## |      8|   320|     0.3|
## |      7|    21|     0.0|
## |      4|    10|     0.0|
##
##
## ## discharge_disposition_id
##
## | Level| Count| Percent|
## |-----:|-----:|-----:|
## |      1| 60234|    59.2|
## |      3| 13954|    13.7|
## |      6| 12902|    12.7|
## |     18|  3691|     3.6|
## |      2|  2128|     2.1|
## |     22|  1993|     2.0|
## |     11|  1642|     1.6|
## |      5|  1184|     1.2|
## |     25|   989|     1.0|
## |      4|   815|     0.8|
## |      7|   623|     0.6|
## |     23|   412|     0.4|
## |     13|   399|     0.4|
## |     14|   372|     0.4|
## |     28|   139|     0.1|
## |      8|   108|     0.1|
## |     15|    63|     0.1|
## |     24|    48|     0.0|
## |      9|    21|     0.0|
## |     17|    14|     0.0|
## |     16|    11|     0.0|
## |     19|     8|     0.0|
## |     10|     6|     0.0|
## |     27|     5|     0.0|
## |     12|     3|     0.0|
## |     20|     2|     0.0|
##
##
## ## admission_source_id
##
## | Level| Count| Percent|
## |-----:|-----:|-----:|
## |      7| 57494|    56.5|
## |      1| 29565|    29.1|
## |     17|  6781|     6.7|
## |      4|  3187|     3.1|
## |      6|  2264|     2.2|
## |      2|  1104|     1.1|

```

```

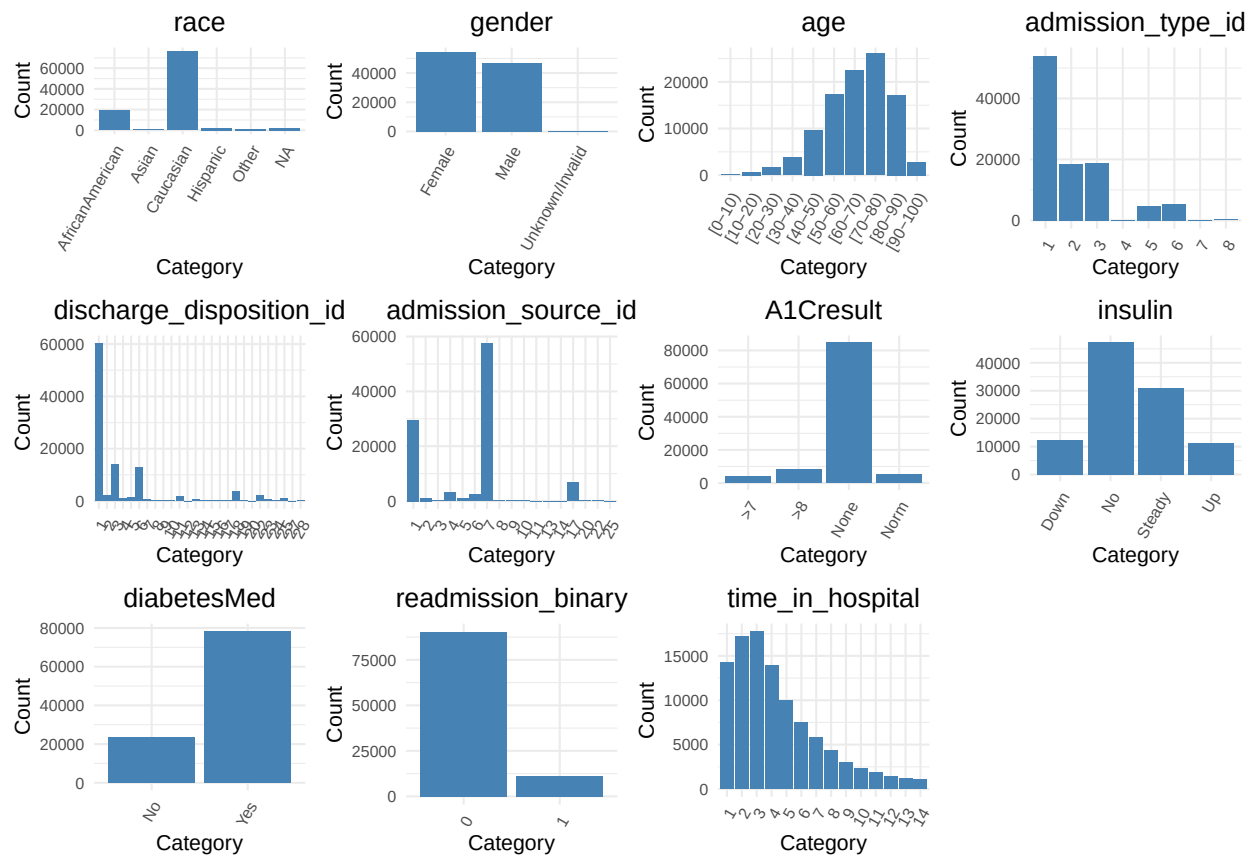
## |      5|    855|    0.8|
## |      3|    187|    0.2|
## |     20|    161|    0.2|
## |      9|    125|    0.1|
## |      8|     16|    0.0|
## |     22|     12|    0.0|
## |     10|      8|    0.0|
## |     11|      2|    0.0|
## |     14|      2|    0.0|
## |     25|      2|    0.0|
## |     13|      1|    0.0|
##
##
## ## A1Cresult
##
## |Level | Count| Percent|
## |:-----|-----:|-----:|
## |None  | 84748|    83.3|
## |>8    |  8216|     8.1|
## |Norm  |  4990|     4.9|
## |>7    |  3812|     3.7|
##
##
## ## insulin
##
## |Level | Count| Percent|
## |:-----|-----:|-----:|
## |No     | 47383|    46.6|
## |Steady | 30849|    30.3|
## |Down   | 12218|    12.0|
## |Up     | 11316|    11.1|
##
##
## ## diabetesMed
##
## |Level | Count| Percent|
## |:-----|-----:|-----:|
## |Yes    | 78363|     77|
## |No     | 23403|     23|
##
##
## ## readmission_binary
##
## | Level| Count| Percent|
## |-----:|-----:|-----:|
## |      0| 90409|    88.8|
## |      1| 11357|    11.2|
##
##
## ## time_in_hospital
##
## | Level| Count| Percent|
## |-----:|-----:|-----:|
## |      3| 17756|    17.4|

```

##		2		17224		16.9	
##		1		14208		14.0	
##		4		13924		13.7	
##		5		9966		9.8	
##		6		7539		7.4	
##		7		5859		5.8	
##		8		4391		4.3	
##		9		3002		2.9	
##		10		2342		2.3	
##		11		1855		1.8	
##		12		1448		1.4	
##		13		1210		1.2	
##		14		1042		1.0	

```
#plots
```

```
gridExtra::grid.arrange(grobs = plots, nrow = 3, ncol = 4)
```



Model Development

Posterior Inference and Model evaluation

Interpretation and Reporting